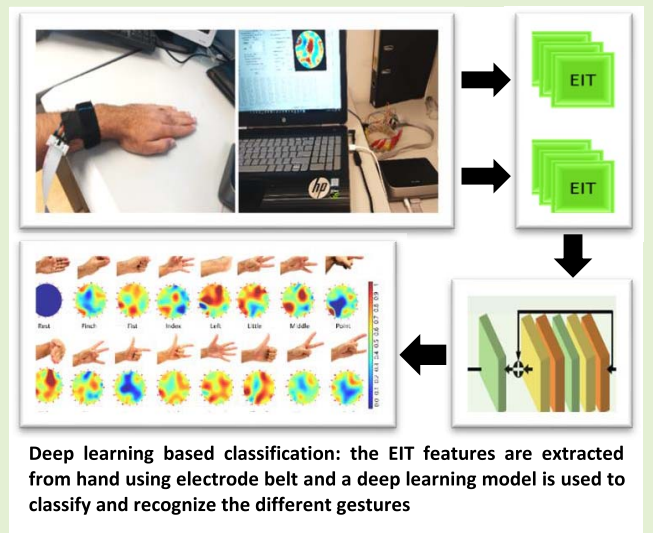


Hand Gestures Classification Using Electrical Impedance Tomography Images

Mehmood Nawaz^{ID}, Russell W. Chan, Anju Malik, *Member, IEEE*, Tariq Khan, and Peng Cao^{ID}

Abstract—Human–computer communication using hand gestures has always been difficult. More than half a century ago, people used different ways of interaction with computers from the early mediums such as perforated game cards. Nowadays, if a richer lexicon of gestures is given, people can communicate more effectively with computers. Machine learning is now used to recognize and classify hand gestures in a more precise way. In order to increase the communication between computers and humans, we proposed a technique, which uses a wearable low-cost device to generate the electrical impedance tomography (EIT) images to recover the inner impedance structure of a user's wrist. This is done by measuring the transverse impedance between all the 16 pairs of electrodes of wrist band that lie on the skin of the user hand. The proposed technique is enough to integrate the technology into the prototype wrist band to monitor and classify gestures in real time. We have conducted a study of 16 gestures with a focus on gross hand and pinch finger gestures. The results evaluation shows that the gross hand gestures achieved 90% accuracy in wrist position, while pinch gestures achieved 93% accuracy.

Index Terms—Bioimpedance, electrical impedance tomography (EIT), gesture classification, high-resolution image, image reconstruction, wrist band.



I. INTRODUCTION

HAND gesture recognition is considered an important communication tool, which is used in several human–computer interaction (HCI) scenarios, such as gaming [1], sign language translation [2], and voice generation for speech impairments [3]. The hand gesture recognition pipeline mainly consists of two blocks: first is the hand gesture capturing block, where the gesture is captured and stored in computer memory, and second is the gesture recognition block

that uses machine-learning techniques to classify the given gesture.

One of the commonly used methods for gesture capturing is the camera-assisted technique [4], [5], where an image of a single gesture is taken and given as an input to the recognition block [6], [7]. It is a traditional approach and has several functional drawbacks, i.e., the need for line-of-sight and application restrictions, such as the inability to develop a prosthetic hand. A recent tomography-based gesture capturing technique is proposed, which is an efficient medical imaging technique that uses impedance and low amplitude current for cross-sectional analysis of the composition and inner structures of the conductive object [8]. Electrical impedance tomography (EIT) is a particular case of tomography, which estimates the impedance distribution from different pairs of electrodes surrounding an object [9]. When a hand gesture is produced, a temporal difference in conductivity distribution is formed by the contraction/relaxation of muscles and bones under the hand skin that can be determined by the EIT system.

The benefits of the EIT system include a radiation-free environment, rapid reaction time, and nondestructive nature. The EIT technology has been applied in several diverse fields

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of study, such as biotechnology [10]–[12], material engineering [13], medical examination [14], [15], and civil engineering [16]. In addition, EIT is noninvasive and comparatively inexpensive to other tomographic techniques, such as magnetic resonance imaging (MRI) and computed tomography (CT) images.

For real-time gesture recognition, the resolution of the generated EIT image should be high. In order to obtain high-resolution images, the commonly employed approach is to increase the electrode count [17], [18]; more electrodes allow us to collect more sophisticated impedance distribution, which helps to reconstruct a high-resolution image. The large number of electrodes increases the EIT image resolution as well as computational power of the system. Silvera-Tawil *et al.* [19] proposed a technique that is used to reconstruct the EIT image within the conducting domain. This technique is not good enough for high-resolution images in EIT [18]. The low-resolution EIT image is not suitable for the medical domain, where noninvasive and nondestructive operations are needed.

An EIT-based gesture recognition approach [20] is used to optimize the drive pattern of impedance and decoding grasp strength [8] for EIT images. In these studies [8], [20], the 2-D EIT projection systems are used to project the 3-D muscular convulsion with low spatial resolution on a cross-sectional plane. These approaches [8], [20] are vulnerable to error detection. In addition, in [20], the experiments were performed only for static gestures. Similarly, the drive patterns [21] were optimized in the neighborhood by repositioning the electrodes in the EIT system. The EIT images of the nerve [22] were analyzed, and a two-cylinder finite-element (FE) model was constructed that used 16 electrodes. All of these studies [8], [20]–[22] attempt to produce high-resolution EIT images with high computational power that is insufficient for an ideal gestures classification system.

An armband-based EIT approach is proposed to interpret the hand movement [23], which is called “TOMO,” derived from the tomography. This approach first captures the generated impedance signal and then uses a machine-learning approach to classify the given gesture. This approach uses a two-terminal measurement method with eight electrodes. Later, the idea of TOMO was further improved by using 16 electrodes with a complex measurement scheme of four terminals [24]. These studies show that the quality of the obtained EIT images can be improved by increasing the number of electrode measurements. However, the increase in electrode measurements raises the difficulties in multifaceted data acquisition and cost processing time. The existing EIT-based gesture recognition approaches [18]–[23] are insufficient to obtain high-resolution images at low cost. However, an efficient image reconstruction and classification approach is needed to classify the hand gestures in real-time applications.

In this article, we propose a robust deep-learning-based approach for hand gesture classification and recognition. The proposed integration module integrates the initial EIT images to obtain a high-resolution EIT image, and the localization process identifies the region of interest (ROI) in generated

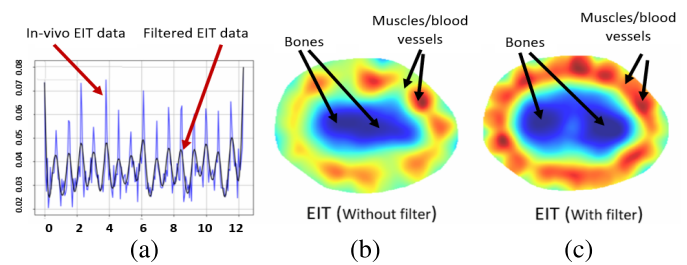


Fig. 1. Effects of bandpass filter on EIT data. (a) 1-D filtered and unfiltered signal. (b) 2-D EIT image without filtering data. (c) 2-D EIT image with filtering data. Thus, the filtered EIT data have high features, including clear bones and muscles area of the wrist.

impedance data of the wrist tissues using a bandpass filter, as shown in Fig. 1. The initial EIT images are obtained by a fully wearable EIT-based wristband equipped with a 16 electrodes band. The localization process highlights the muscle's features. It smooths the irregular behavior of the *in vivo* wrist data signal, as shown in Fig. 1. In addition, we proposed a Bayesian-ResNet-18 classification technique to classify and recognize the different hand gestures.

Contributions: The main contributions of this article are given as follows.

- 1) A consistence image reconstruction technique is used to obtain a high-resolution EIT image by integrating initial EIT images that are generated by the wrist band with 16 electrodes.
- 2) The localization process is presented to extract ROI from EIT images using the bandpass filtering technique. It smooths the raw data signals and highlights the geometrical features of the wrist tissues.
- 3) A deep-learning-based classification approach is introduced for gesture recognition and classification. It uses ResNet-18 and Bayesian optimizer as a backbone.
- 4) Experimental results on a different number of subjects validate the efficacy of our classification system and show significant improvement in quality.

The rest of this article is organized as follows. The literature review is discussed in Section II. The EIT image reconstruction is explained in Section III. The wrist data acquisition and classification is shown in Section IV. The experimental results are described in Section V, and Section VI concludes the proposed technique.

II. LITERATURE REVIEW

Hand gestures are seen as an essential interaction tool to convey information. Hand gestures are in use even before the invention of language [26]. In the modern technology, hand gesture recognition systems have a wide area of applications in HCIs [27], such as sign language detection [24] and robotic arms for amputees [28]. In general, most of the approaches for gesture recognition are based on camera assistance and wearable devices. The camera assistance-based approaches discussed in [5], [29], and [30] are flexible and do not require any wearable devices. However, these approaches are not applicable where the interpretation of the fine details of the muscular movement is essential. They did not give the inner

geometry of wrist hand muscles during different gestures movement.

The commonly used technique for gesture recognition in wearable devices is surface electromyography (SEMG) [31], [32], where muscular activities are recorded by attaching electrodes to the skin surface. This approach is simple and noninvasive. The performance comparison of different SEMG approaches for forearm and wrist level is presented in [33]. This study [33] demonstrated that the high-quality SEMG signal with less noise could be obtained from the wrist position compared to the forearm. The reconstructed images of SEMG approach are low resolution [34]. In addition, SEMG can detect signals at a maximum of 1 cm beneath the skin, while signals from deeper muscles are not detectable. Other approaches, such as implantable myoelectric sensors (IMESs) [35] and transverse intrafascicular multichannel electrode [34], are invasive and have limited applications [20] than the EIT system. Wu *et al.* [36] proposed a human-machine interface that connects the user to a hand prosthesis. It successfully employs an array of electrodes enclosed on a wristband on the user's forearm to perform EIT. Total eight electrodes are used to measure the bioimpedance of hand tissues. Despite the fact that training just takes a few minute, which is the drawback of this system, the early investigation in this system shows that the feature changes can be acquired due to the electrode band location.

Electrical impedance tomography (EIT) is considered a possible alternative technique for signal acquisition. In [37] and [38], the viability of EIT is studied for wearable devices to detect hand motion, specifically, touching one finger with the other. These studies have limited use in two-handed gesture touching applications. The estimation of wrist angle for gesture recognition applications is investigated in [39]. A comparative study for intraelectrode distance in the EIT system is presented in [40], where the optimal distance is defined by estimating the change in real and imaginary parts of impedance measurements.

A study related to flexible EIT-based sensor for fingertip movement detection is reported in [41]; this approach is based on eight and 16 electrodes and uses a baseline update procedure called iSoft. The use of opposite drive pattern is proposed in [42] to identify the contact location for robotic application; this technique is based on EIT of eight and 16 electrodes with supervised quadratic classification algorithms. This approach achieves better performance compared to flexible EIT. However, this study did not consider the optimization of data acquisition methods. Other EIT-based approaches [22]–[24] used additional use of electrodes to enhance the image quality, which is a highly nonlinear problem. The conventional approaches adopt finite-element methods (FEMs), which is resulting in high computational cost [19]. Some techniques are using low computational machine-learning algorithms for object segmentation and detection in images [43]–[50]. Thus, the high-quality image acquiring on low computational cost is a main challenge in the EIT system.

Our body tissues have different impedance values. For example, muscles and blood veins have low impedance, while skin and bones tissues are high in impedance, as shown

in Fig. 2. The most common approaches use a specific frequency range to measure the bioimpedance. Some instruments use multiple frequencies such as multifrequency bioelectrical impedance analysis (MF-BIA). The most basic instruments are based on a single-frequency measurement of the bioelectrical impedance (SF-BIA) or bioimpedance spectroscopy (BIS).

There are also several methods for assessing the bioimpedance outcomes. Bioelectrical impedance vector analysis and real-time analysis are the most significant. The muscles and blood have comparable resistance values and their impedance is not significantly important if we increase the frequency up to 350 kHz. It is proven that it cannot be supposed in impedance plethysmography. In a different research, the researchers have used below 10-kHz frequency to achieve a constant blood and tissue impedance ratio. The impact of polarization processes of electrodes is examined on impedance measurement devices. The hand tissue resistance and capacity over a broad range of frequencies are described in Fig. 2. The different conductivity values of body tissues are very helpful to reconstruct an instructive EIT image.

III. EIT APPROACH

EIT is a noninvasive imaging technique, which aims to recover the conductivity of muscles in ROI from the surface using electrical measurements. A typical 2-D wrist-shaped domain is used for EIT data. Data acquisition and EIT image reconstruction of different wrist tissues are discussed in the following.

A. Electrical Impedance of Body Tissues

Different approaches are using bioimpedance or biocapacitive sensing technique in medical applications, such as computerized scales with body fat detection capabilities, and accurately capture and analyze the human movement [9], [22]. The research area of hand gestures classification is based on bioimpedance that is calculated by the Gauss-Newton (GN) method as discussed in [20]. These types of systems are equipped with a single transmitter and receiver, which can operate on a single or multiple frequency with multiple electrode patterns (four, eight, and 16 electrodes) to get a consistence bioimpedance value. The multiple electrode-based technique has been used in different approaches, such as Tran *et al.* [6], which examines cross-sectional impedance among all four and eight pairs of electrodes on the forearm of a user. This technique is limited to eight electrodes to recognize the gestures. Tan *et al.* [5] proposed a neckband to test user neck impedance. The neck-related actions, such as head movements, talk, coughing, and swallowing, can be determined by this technique. This technique has a similar idea to our approach but includes high computational cost in real-time features extraction.

B. EIT Image Reconstruction

To generate the 2-D interior image, we used the GN reconstruction approach, one of the most popular reconstruction algorithms for industrial applications. Compared to other algorithms, it has low computational cost and is widely used

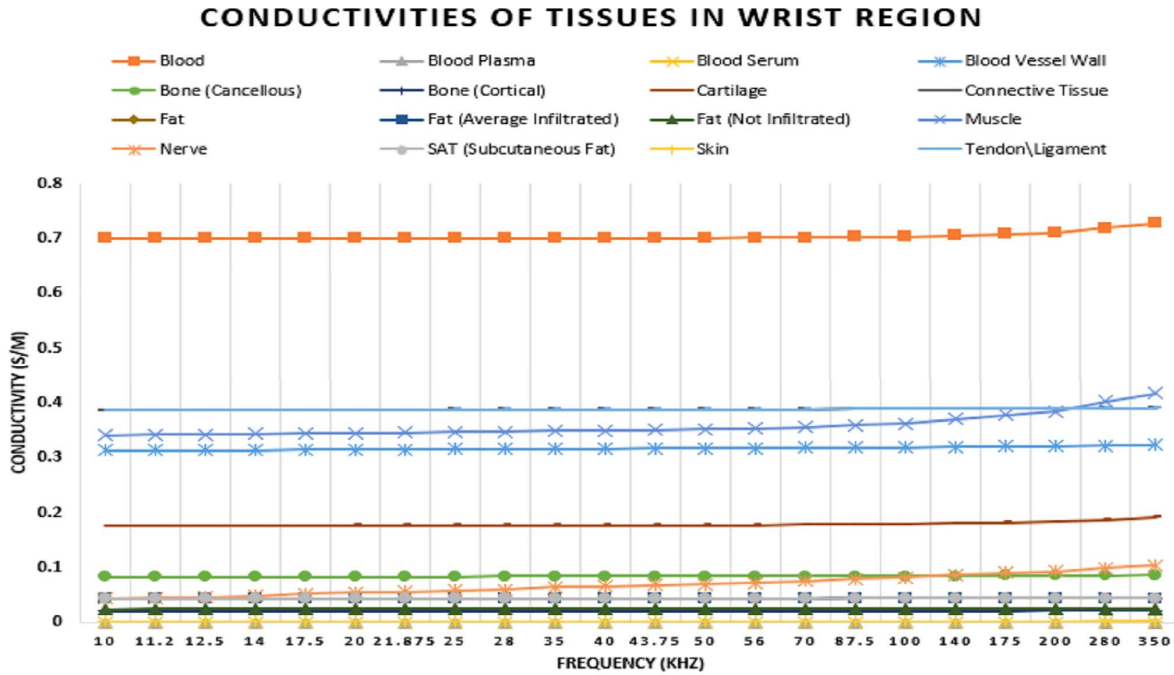


Fig. 2. Conductivity values of different body tissues in the wrist region. From left to right, the frequency range increases from 10 to 350 kHz, but the conductivity of tissues remains constant [25]. The vertical values show conductivity (s/m) and the horizontal values show the frequency range (kHz).

for real-time image reconstruction. Many other image reconstruction techniques are discussed in [7].

Reconstruction of images in the EIT system is a nonlinear issue, and however, linear approximations are highly helpful. The presence of measurement noise and geometric uncertainty in clinical experiments affects the image reconstruction efficiency in EIT data. The linear algorithms also offer the advantage of image reconstruction, where the impacts of data artifacts can be detected easily. Thus, the linear reconstruction might be realized as a quick matrix multiplication. The linear EIT image reconstruction is represented by a matrix R that translates the measurements (y) to a reconstructed image (X)

$$X = R \times y. \quad (1)$$

This section discusses the GN EIT image reconstruction technique, which has been used frequently from the last decade for EIT systems. This technique enables the adoption of advanced regularized models of the inverse EIT issue, which may then express the answer as a linear reconstruction matrix that enables fast real-time image reconstruction. The Graz consensus reconstruction algorithm for EIT (GREIT) contains the FE models and has a systematic approach to optimize the linear reconstruction matrix in different clinical systems. It was proposed to reconstruct the lung images in [51]. It mostly supports the minimum size of the dataset, as discussed in [51]. It might be a dilemma in EIT image reconstruction for large dataset training, while our proposed approach has been focused on the features of the EIT image using the GN technique with low computational cost. The connection between (x) and (y) can be linearized for modest changes to the reference

conductivity (r)

$$Y = J \times X + n \quad (2)$$

where $J \in \mathbb{R}^{N_m \times N_n}$ shows the sensitivity matrix, X shows the reconstructed image, and n shows the measurement noise that is assumed a white Gaussian noise. J is calculated by the finite-element analysis (FEM), which depends on the reference conductivity, injection current, and electrode model (EM)

$$[J] = \frac{\gamma[Y]\sigma_n}{\gamma[X]\sigma_m} \quad (3)$$

where σ_n and σ_m are the foreground and background conductivity, respectively. This system is undetermined because $\sigma_n > \sigma_m$. Regularization methods are necessary to compute a conductivity change estimate x , which is true to both the measurements y and *a priori* reconstruction on an image. The GN uses a generalized Tikhonov regularization that is the minimum of the sum of a quadratic norm for inverse problem solution x'

$$\|y - Jx'\|_{\sum_n^{-1}}^2 + \|x - \bar{x}\|_{\sum_x^{-1}}^2 \quad (4)$$

where $x - \bar{x}$ is the change of conductivity in different EIT images. $\sum_n^{-1}$ is the covariance matrix of measurement noise n . $\sum_x^{-1}$ is a diagonal matrix due to uncorrelated measurement noise n with $\sum_x^{-1} = \sigma_i^2$, where σ is noise variance at measurement i . The expected covariance of the image is denoted by $\sum_x \in \mathbb{R}^{N_N \times N_N}$. Covariance matrices $\sum_n$ and $\sum_x$ are often not computed directly but are heuristically modeled from *a priori* assumptions as $V = 2$, where n is the average measurement noise amplitude, x is the *a priori* amplitude of conductivity change, and V represents the measuring precision

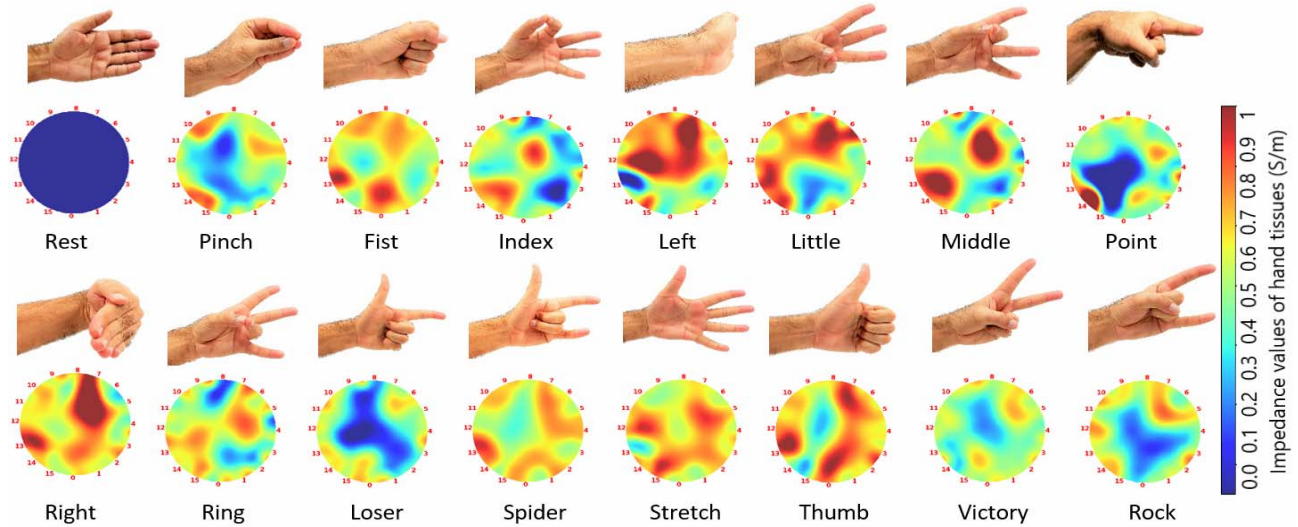


Fig. 3. Electrical impedance features of different gestures at wrist position. There are 16 different gestures with EIT feature maps. The right-side bar represents the impedance values from 0 to 1 (best viewed in color).

[see (5) and (6)]. Each diagonal element in uncorrelated noise is proportional to the signal-to-noise ratio and ($V = I$) for different EITs with identical channels. The regularization matrix P can be used to represent the probability of image components and their interactions. The one-step invariance obtained as

$$X = \left(J^T \frac{1}{\sigma_n^2} V^{-1} J + \frac{1}{\sigma_n^2} P^{-1} \right)^{-1} J^T \frac{1}{\sigma_n^2} V^{-1} y \quad (5)$$

$$X = \left(J^T V^{-1} J + \lambda^2 P^{-1} \right)^{-1} J^T V^{-1} y \quad (6)$$

where λ is the regularization hyperparameter, which controls the tradeoff between the noise attenuation and resolution in the reconstructed image. The matrix $R = (J^T V^{-1} J + \lambda^2 P^{-1})^{-1} J^T V^{-1}$ is the linear matrix of one-step inverse function. R can be reconstructed in the data form or Wiener filter form to decrease computing time and enhance inverse accuracy and stability, which is calculated as

$$R = P J^T \left(J P J^T + \lambda^2 V \right)^{-1} \quad (7)$$

where P becomes an identity matrix and employs the zero-order Tikhonov regularization if image elements are considered to be independent with identical variance. Because measured data are significantly more sensitive to boundary image components, such solutions tend to shift reconstructed noise toward the border for EIT. P can be scaled in accordance with the sensitivity of each element. The obtained EIT raw data contain materials and skin contact noise that is removed by a bandpass filter, as shown in Fig. 1. Finally, we get the 2-D EIT image, which is the difference of reference (rest position gesture) and current hand gesture in EIT that is shown in Fig. 3.

IV. SYSTEM IMPLEMENTATION

We used a biosensor board and 16-electrode-based wrist band that are manufactured by Gense Technologies Ltd. [52]

to collect the hand gestures data. The obtained data include biotissues' conductivity values that are used to reconstruct the EIT image. The deep-learning-based classifier is used to classify the hand gestures (discussed in Section IV).

A. Wrist Band and EIT Device

The 16-electrode-based belt is used in the left hand, as shown in Fig. 5(a). It is made by 16 electrodes that are equally spread, as shown in Fig. 5(a). In order to establish a strong contact between the electrodes and the skin of users, the belt elasticity has been adopted. This belt is flexible and easily adjustable to any size of the arm.

The EIT device is a beta version that is built by Gense Technologies Ltd., as shown in Fig. 5(b). This device has a build-in analog-to-digital converter that enables impedance measurement with a precision of 0.1 Hz at a specified frequency range between 10 and 350 kHz. We have used a 140-kHz excitement signal, which shows the consistence impedance values of the body tissues during the muscle movements.

V. WRIST DATA ACQUISITION

A. Experimental Study

We recruited 22 people with a mean age of 28, and all are right-handed. The research was completed in two parts and took around 2 h. Since all our participants were right-handed, the system was worn on the left wrist position (watch position).

B. Data Acquisition Procedure

Our investigation included two data collecting phases that served various experimental goals. Phase one data are used to measure user precision, required training volume, and user accuracy (i.e., universality). Phase two data are used for the reproducibility assessment. All phases data are taken after a short introduction about wrist protocols. All subjects wore

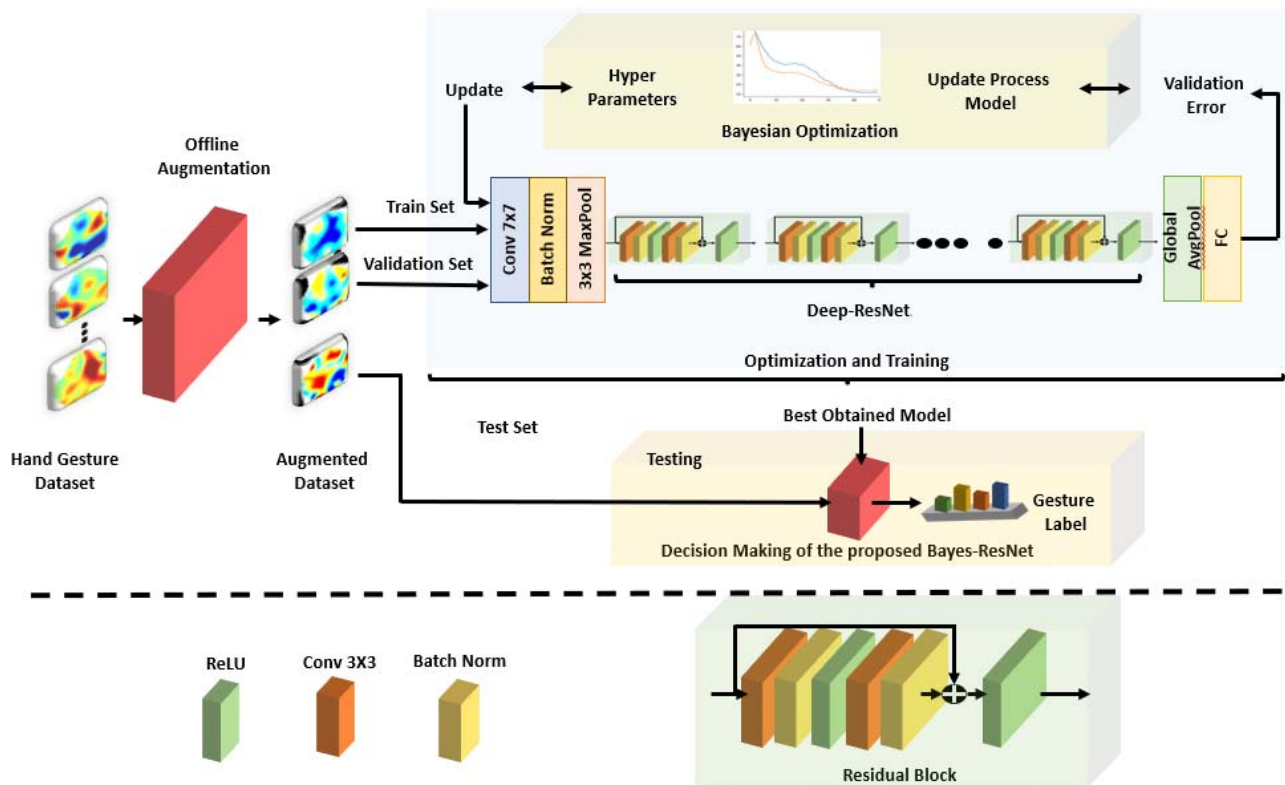


Fig. 4. Flowchart of gestures classification model based on ResNet. It follows the VGG model design with the same number of output channels. Each convolution layer has ReLU activation and batch normalization function. This kind of classifier required the same shape of output and input layers. An augmented dataset of EIT hand gestures is used to train this classifier.

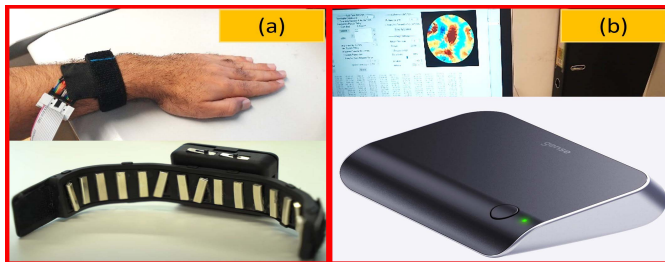


Fig. 5. EIT data acquisition setup. (a) Wrist band position in left hand. (b) EIT setup with Gense devise. This device reconstructs the 1-D impedance signal to the 2-D image as shown on the laptop screen. It is a real-time EIT image reconstruction.

a wrist band in their left hand. Our method requires conductive gel such as other electromyography (EMG) devices. Participants were then instructed to do one gesture at a time. The gestures are retained by the participants until the laptop released a beep, which took a few seconds to collect different features. During data collection, participants received feedback (e.g., graphical) which shows the conductivity of electrodes with hand skin. During phase one, participants do 16 gestures one by one with 20-s break among each gesture. After all gestures, a single cycle of data collecting is completed. We can acquire data based on frequency cycle or time-based. However, in all experiments, we adopted a time-based strategy up to 60 s.

During phase two, a 1-min break was then granted to participants to remove the wrist band. After the break, the band is reinstalled on the wrist (as close as feasible to the original

place at hand). Participants performed the same gestures as previously.

C. Gestures Sets

We divided a set of gestures into two subsets such as hand gestures and pinch gestures. The hand gesture data contain rest, left, right, point, thumb, stretch, and fist gestures, as shown in the first row of Fig. 6. The pinch gesture data contain finger gestures, such as pinch, index pinch, middle pinch, ring pinch, little pinch, loser pinch, spider pinch, victory pinch, and rock pinch, as shown in the second row of Fig. 6. We represented a relaxation gesture as a neutral gesture in both datasets. Thus, the total number of pinch and hand gestures is 16 in our proposed technique.

D. Gesture Classification

The hand gesture dataset is classified by using the Bayesian-ResNet-18-based convolution neural network. It is a Python-based framework that uses the Bayesian optimizer to enhance the computing capacity of the proposed method. The ResNet-18 CNN model includes certain pretrained weights that may be used immediately for feature extraction, prediction, and fine-tuning.

In the gesture classification, we used a dataset that includes different hand gestures images, such as down, up, stretch, victory, pinch, fist, left, right, little, ring, index, middle, spider, thumb, point, and rock. The Bayesian-ResNet-18 model is fast and straightforward that supports the proposed framework on both CPUs and GPUs smoothly.

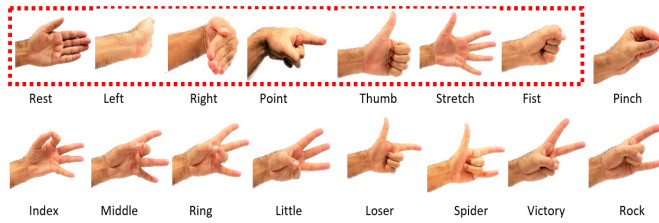


Fig. 6. Gestures dataset: images in red dotted lines are simple hand gestures and other images are pinch gestures.

E. Bayesian-ResNet-18 Robust Classifier

Fig. 4 presents the general design of gesture classifier model based on the Bayesian and ResNet-18 model. The proposed classifier has three major phases, such as offline data augmentation in the raw dataset, training of the Bayesian ResNet-18 model for optimization, and the decision-making network. In the first phase, the offline data augmentation technique is used on EIT images due to uneven distribution samples. This method is preferred for smaller classes with fewer number of samples to increase the size of the classes by a transformation factor, resizing, rotation, and so on. The augmented dataset is split into three subsets as training, validation, and testing. The second phase contains training and optimization of the proposed classifier. This phase uses training and validation datasets as an input [53]. For the test phase, the decision-making network uses a specific test dataset to check the efficiency of the model.

Our dataset is based on reconstructed EIT images that are different from the ImageNet dataset. A good dataset is always helpful for a classifier to smooth its training. The optimal decision-making part contains the Bayesian optimizer, which is used to differentiate the gestures, as shown in Fig. 4. For updating the optimization process, a validation error is used.

1) *ResNet-18 Architecture*: ResNet is based on the VGG-16 model, which uses 3×3 convolutional layers. The residual block is made up of two 3×3 convolutional layers that all have the same number of output channels. Each convolutional layer is followed by a batch normalizer layer and a ReLU activation module. The input is then inserted just before the final ReLU activation function, bypassing the two convolution phases in the process. In order to be combined together in this architecture, the outputs of the two convolutional layers must have the same form as the input. If the number of channels is modified, a new 1×1 convolutional layer is needed to turn the input into the correct shape for the addition operation. Fig. 7 shows two different sorts of networks, with and without 1×1 convolution.

The ResNet-18 model contains the same first two layers as the GoogLeNet, which includes a 7×7 convolutional layer with 64 output channels and a stride of 2, followed by a 3×3 maximum pooling layer. The distinction between ResNet and other convolutional networks is the batch normalization layer that is added after each convolutional layer. GoogLeNet is composed of four modules, each of which contains inception blocks. ResNet employs four modules, each with the same

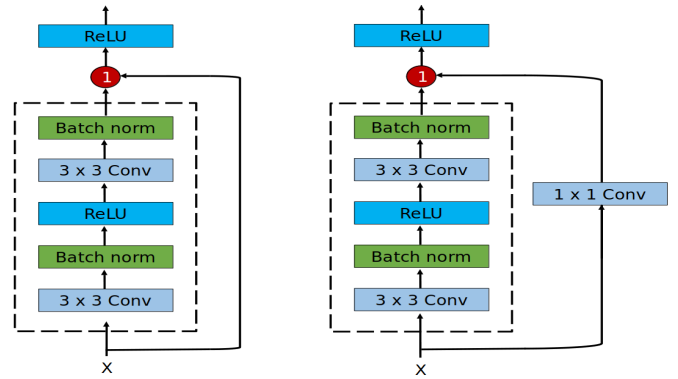


Fig. 7. ResNet block with and without convolution (1×1 CONV).

number of output channels, which are made up of residual blocks. The number of channels in the first module is equal to the number of channels in the input. It is not essential to reduce the height and width because a maximum pooling layer with a stride of 2 has already been applied. The number of channels in the first residual block of each succeeding module is doubled when compared to the previous module, and the height and width are halved.

Note that the first module has undergone special processing. After that, we load ResNet with all of the modules. Each module in this example has two residual blocks. Finally, we add a global average-pooling layer, similar to GoogLeNet, followed by the output of the fully connected layer. Each module consists of four convolutional layers (excluding the 1×1 convolutional layer). There are 18 layers in total, including the first 7×7 convolutional layer and the last fully connected layer. As a result, this model is called ResNet-18. Different ResNet models, such as the deeper 152-layer model, can be created by different numbers of channels and residual blocks. ResNet has a similar architecture to GoogLeNet. ResNet's structure is more straightforward and easier to change. As shown in Figs. 4 and 7, all of these factors contributed to ResNet's rapid and widespread adoption.

2) *Bayesian Optimization*: In both machine and deep-learning algorithms, hyperparameters play a major role since they closely manage the actions of training algorithms and greatly impact the models' performance. Therefore, it is a significant task to optimize hyperparameters, particularly in deep learning of medical images. In general, there are two types of optimized parameters searching such as manual and automatic. Most of the models suggest manual searching for hyperparameters, which is insufficient in the big dataset [30], [37].

The tuning of the hyperparameters is a problem in the optimizing process. A Bayesian optimization is an effective approach to solve these kinds of problems [30], [39]. The Bayesian optimization is based on a usual approximation, which is similar to the idea of the likelihood of posterior. Here, observations of the algorithm are derived from the model assessments, where the online learning results are presented. This means that our network requires a training process in the Bayesian optimization. This model traces a function

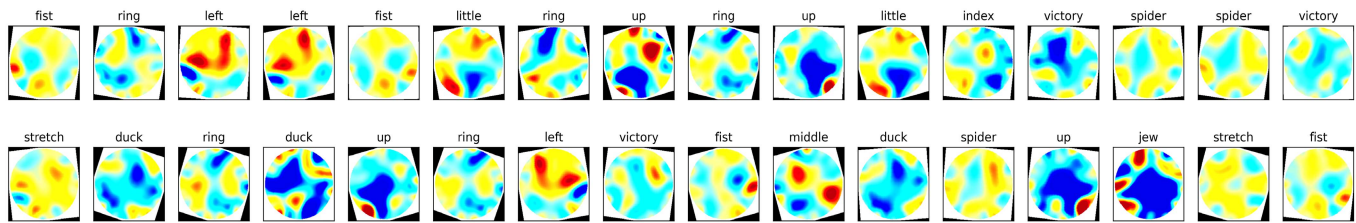


Fig. 8. Data augmentation samples of 16 different gestures. We applied six different types of data augmentation (contrast, resize, rescaling, illumination, cropping, and brightness), best viewed in color.

throughout the training, which we know exclusively from the learning data. The major goal is to achieve the hyperparameters, which makes a learning outline maximum [40] during the Bayesian optimization stage.

VI. EXPERIMENTAL RESULTS

We have carried out detailed experiments on 22 different subjects' data. This section includes the data training, parameters settings, reproducibility of gestures, and universality of gestures. The qualitative detail of these experiment results is discussed in the following.

A. Offline Augmentation

The imbalance ratio of the class distribution of the dataset has a big effect on the classification of both traditional machine-learning and deep-learning methods performance. In the study, Zheng *et al.* [39] investigated systematically how data imbalances affect the performance of CNN classification. The results of the investigation show a destructive influence on the classification performance due to imbalance class distribution. We execute a comprehensive offline augmentation over gestures images to deal with this unfavorable condition. First, by flipping each image, we acquire the reflected version of the original images. Then, both the original and flipped images are used in data augmentation. It is shown in Fig. 8.

B. Training Data

We have divided gesture data into three sets: training data (80%), validation data (10%), and test data (10%). The training dataset contains 1500 images with labels, the validation dataset contains 200 images, and the test dataset contains 200 images with 16 different classes. The hand gestures images have size 256×256 with PNG format. The loss and accuracy of training and validation dataset are shown in Fig. 10.

C. Reproducibility of Data

The ideal hand gestures classification system should be stable over time. The proposed method is stable and smooth during retraining. To check the stability of our method, we acquiesce the data from different users. The stability of model also depends on the structure similarity of the EIT image. We also checked the SSIM index of different subjects' data. We have trained our model on one subject dataset and test on other subjects datasets.

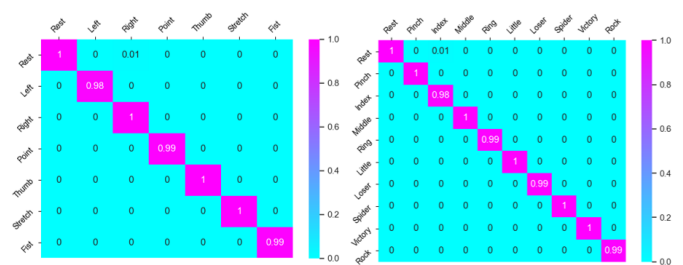


Fig. 9. Classification and confusion matrix of 16 different gestures. (a) Accuracy value of six different gestures. (b) Confusion matrix based on precision values of each gestures. Our classifier gives the good results on 16 different gestures.

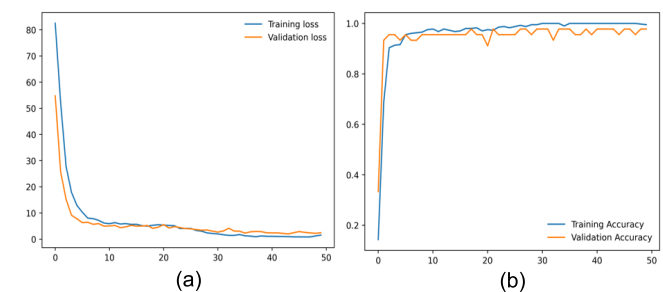


Fig. 10. Training and validation loss and accuracy. (a) Classifier training and validation loss of gestures dataset. (b) Accuracy of classifier on training and validation datasets. The minimum loss and high accuracy show the efficiency of the proposed classifier on 16 different gestures (best viewed in color).

D. Universality of Data

The universality of the proposed method is also checked on the cross users' datasets. For this purpose, we used the combination of different subjects' data and test it on the single subject data. The confusion matrix in Fig. 9 shows the inner gestures similarity score. We have divided our gesture set into two sets. Fig. 9(a) shows the confusion matrix of rest, left, right, point, thumb, stretch, and fist and Fig. 9(b) shows the confusion matrix of rest, pinch, index, middle, ring, little, loser, spider, victory, and rock. The first set of gestures is widely used and the second set of gestures is rarely used in applications. The proposed method has 90% accuracy in the first gestures set and 93% accuracy in the second gestures set. We have also checked the cross user reproducibility score, which is shown in Fig. 11. The quantitative analysis of each hand gesture with respect to precision, recall, and F-measure is shown in Fig. 12. These results show that our proposed method performs well on each hand gesture.

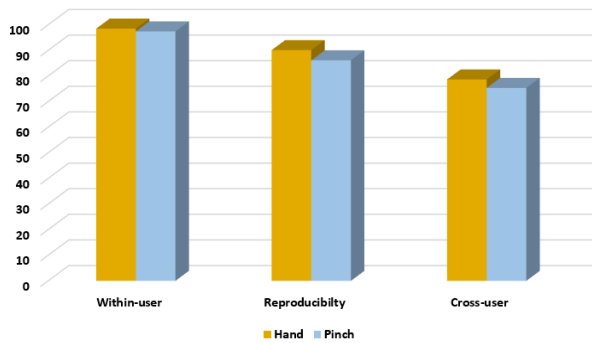


Fig. 11. Classification accuracy of hand gestures and pinch gestures with respect to re-reproducibility, within user and cross user. The hand gestures classification accuracy is little high compared to pinch. During the pinch, the conductivity change is high than hand gesture change.

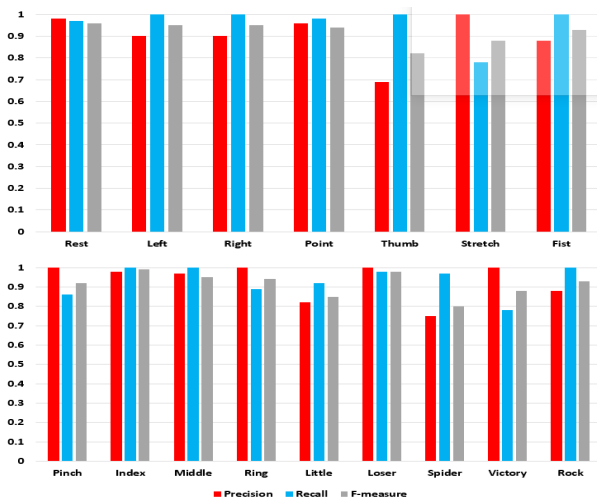


Fig. 12. Quantitative analysis of different hand gestures with respect to precision, recall, and F-measure score (best viewed in color).

E. EIT Wrist Applications

We can deploy our proposed system in different smart wearables in the future. We can also adopt different EMs on wrist band to collect the electrical impedance features (4, 8, 16, ...). By wearing wrist band device, the user can easily recognize the gestures of the hand. Zhang and Harrison [54] also proposed the similar device for smartwatches, but it has limited features due to less number of electrodes (eight electrodes) and used simple machine-learning features to classify the hand gestures with high computational cost. We ultimately intend to include this technique in future smart wearables that can work together collaboratively to support the interactive tasks on small screens with hand gestures and direct touch manipulation.

VII. CONCLUSION

In this article, we have proposed a deep-learning-based gesture classification technique using EIT images. It finds the cross-sectional bioimpedance of hand using 16 electrodes. Our proposed method recovers the impedance distribution of hand tissues. These impedance distributions are used to reconstruct the EIT image. People can communicate even more effectively, if a richer lexicon of gestures is given.

Deep learning is now used to recognize the hand gestures that was always more difficult in past. Deep-learning-based models recognize gestures more precise than ever before. The proposed technique is enough to integrate the technology into the prototype wrist band, to monitor and classify gestures in real time. The proposed method has achieved 90% and 93% accuracy at wrist position using hand and pinch gestures dataset, respectively.

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